

Solving Poisson equation with Successive Overrelaxation method (SOR)

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1st November 2025

1 Introduction

The aim of this project is to solve Poisson equation for the electric potential $V(\vec{r})$ in 2D

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = -\frac{\rho(x, y)}{\epsilon}, \quad (x, y) \in [-L, L] \times [-L, L] \quad (1)$$

The dielectric function ϵ is assumed constant while the charge density ρ is defined below

$$\rho(x, y) = A_\rho x y e^{-(x^2+y^2)} \quad (2)$$

with some amplitude A_ρ to be defined later. Poisson equation must be supplemented with boundary conditions (BC), if it is not stated otherwise we will use the following Dirichlet BC

- left boundary, $x = x_{min}$

$$V(x_{min}, y) = V_{max} \sin\left(\frac{k_1 \pi y}{2L}\right) \quad (3)$$

- top boundary, $y = y_{min}$

$$V(x, y_{max}) = V_{max} \sin\left(\frac{k_2 \pi x}{2L}\right) \quad (4)$$

- right boundary, $x = x_{max}$

$$V(x_{max}, y) = V_{max} \sin\left(\frac{k_3 \pi y}{2L}\right) \quad (5)$$

- bottom boundary, $y = y_{min}$

$$V(x, y_{min}) = V_{max} \sin\left(\frac{k_4 \pi x}{2L}\right) \quad (6)$$

The number factors k_1, k_2, k_3, k_4 are used to change BC between homogeneous (all k_i being zero) and nonhomogeneous one (at least one $k_i \neq 0$), V_{max} is the amplitude of potential at boundary.

1.1 Mesh of nodes and discretization of Poisson equation

To solve Poisson equation we use the simplest approach - the relaxation on mesh of nodes. By first, we shall introduce the mesh, the number of nodes will be the same in both directions i.e. equal N , hence the nodes are equidistant with spatial step $\Delta x = \Delta y = \Delta$

$$\Delta = \frac{2L}{N-1} \quad (7)$$

$$x_i = -L + \Delta \cdot i, \quad i = 0, 1, 2, \dots, N-1 \quad (8)$$

$$y_j = -L + \Delta \cdot j, \quad j = 0, 1, 2, \dots, N-1 \quad (9)$$

From now on we can use potential and density defined on mesh

$$V(x, y) \rightarrow V(x_i, y_j) \rightarrow V_{i,j} \quad (10)$$

$$\rho(x, y) \rightarrow \rho(x_i, y_j) \rightarrow \rho_{i,j} \quad (11)$$

Next we discretize the Poisson equation on mesh by replacing the derivatives with their finite difference approximations. We use 3-point central formula

$$\frac{v_{i+1,j} - 2v_{i,j} + v_{i-1,j}}{\Delta^2} + \frac{v_{i,j+1} - 2v_{i,j} + v_{i,j-1}}{\Delta^2} = -\frac{\rho_{i,j}}{\epsilon} \quad (12)$$

To solve the Poisson equation on mesh we use the relaxation method, therefore the solution at point (i,j) should be extracted from expression 12

$$v_{i,j} = \frac{1}{4} \left(v_{i+1,j} + v_{i-1,j} + v_{i,j+1} + v_{i,j-1} + \frac{\rho_{i,j} \Delta^2}{\epsilon} \right) \quad (13)$$

while in SOR method an additional convergence parameter $\omega \in (0, 2)$ is imposed

$$\text{SOR: } v_{i,j} = (1 - \omega)v_{i,j} + \frac{\omega}{4} \left(v_{i+1,j} + v_{i-1,j} + v_{i,j+1} + v_{i,j-1} + \frac{\rho_{i,j} \Delta^2}{\epsilon} \right) \quad (14)$$

1.2 Stopping criterion

In any iterative method we should apply a stopping criterion which preserves against performing redundant iterations after the solution has already converged. It is common to use electric field energy functional as a marker of stabilization of the potential, mathematically it is expressed as integral

$$S = \int d^2r \left(\frac{1}{2} \vec{E}(\vec{r}) \cdot \vec{E}(\vec{r}) - V(\vec{r}) \frac{\rho(\vec{r})}{\epsilon} \right) \quad (15)$$

where $\vec{E} = -\nabla V(\vec{r})$ is the electric field intensity. Functional S must be calculated for mesh quantities, hence we need two loops to scan entire 2D computational mesh, and the 1-st order derivative is replaced with 2-nd order finite difference formula

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initialize: s=0
for i=1 to N-2 do
  for j=1 to N-2 do
     $E_x = -\frac{v_{i+1,j} - v_{i-1,j}}{2\Delta}$ 
     $E_y = -\frac{v_{i,j+1} - v_{i,j-1}}{2\Delta}$ 
     $s = s + \left( \frac{1}{2}(E_x^2 + E_y^2) - v_{i,j} \frac{\rho_{i,j}}{\epsilon} \right) \Delta^2$ 
  end do
end do
end do

```

Here we skip some information gathered at boundaries, but until the number of boundary nodes are much smaller than these for the inner part of box it seems reasonably justified. If the value of S will change by less than some tolerance threshold TOL in two consecutive iterations (S_k and S_{k+1}) then the relaxation process can be stopped

$$\delta = \left| \frac{S_{k+1} - S_k}{S_k} \right| < TOL \quad \implies \quad STOP \quad (16)$$

1.3 Checking correctness

Although the relaxation method may stop automatically we can not be sure if the potential that has been found is correct. Cheking its correctness is very is simple, one just needs to subtract the left side from the right side in Eq.1 for quantities defined on mesh

$$r_{i,j} = \frac{v_{i+1,j} - 2v_{i,j} + v_{i-1,j}}{\Delta^2} + \frac{v_{i,j+1} - 2v_{i,j} + v_{i,j-1}}{\Delta^2} + \frac{\rho_{i,j}}{\epsilon} \quad (17)$$

This in fact corresponds to calculation of residual vector so in next step its norm usually shall be determined, however we only display the result and look at its amplitude, hence we can also find at which points the local errors are the largest.

1.4 Algorithm for SOR method

In SOR we use only one array to store the potential, the values already upgraded in present iteration are immediately used in calculations at other nodes. Pseudocode for SOR may look as follows

```

initialization :
    L, N, Δ, ε, TOL, Kmax
    vi,j ← 0   i, j = 0, 1, ..., N - 1,
    ρi,j ← Eq.2   i, j = 0, 1, ..., N - 1
    snew,sold = 0

    impose Dirichlet BC if needed for vi,j:
    Eqs. 3, 4, 5, 6

relaxation loop:
for k=1 to Kmax do

    upgrade Neuamann BC if needed for vi,j:

    upgrade solution:
    vi,j ← r.h.s of Eq.14,      for i, j = 1, 2, ..., N - 2

    check the convergence:
    snew ← code in Sec.1.2
    δ =  $\left| \frac{s_{new} - s_{old}}{s_{old}} \right|$ 
    if k>1 and δ < TOL then
        break
    end if
    sold ← snew

end do

```

2 Practical part

Tasks to do

1. Implement the SOR algorithm on computer, then set parameters: $L = 4$, $N = 100$, $\epsilon = 1$, $A_\rho = 0$ (no density source), $k_1 = 1$, $k_2 = -1$, $k_3 = 1$, $k_4 = -1$, $TOL = 10^{-8}$, $K_{max} = 10^4$ and $\omega = 1.5$. Solve the Poisson equation, make 2D maps of the solution and the error (Eq.17).
2. Repeat the calculations as in task 1 but with Neumann BC on the right side

$$\left. \frac{\partial V}{\partial x} \right|_{x=x_{max}} = 0 \quad (18)$$

and the same Dirichlet BC on the rest box's edges as before. By replacing 1-st order derivative with 2-point backward approximation (right boundary) we get condition on mesh

$$\frac{v_{N-1,j} - v_{N-2,j}}{2\Delta} = 0 \implies v_{N-1,j} \leftarrow v_{N-2,j} \text{ for } j = 1, 2, \dots, N-2 \quad (19)$$

Add this BC to SOR algorithm. Solve the problem and make 2D maps of the solution and the error (Eq.17).

3. Now manipulate a little the value of convergence parameter ω . Repeat the calculations as in task 2 for $\omega = 1.0, 1.3, 1.6, 1.9$ (for $\omega = 1$ SOR=Gauss-Seidel). Remember to start each relaxation process with zeroed potential array ($v_{i,j} = 0$), at each iteration save in file the index of present iteration and value of functional S. Make the plots of functional S as function of iteration number for all ω values in one figure. Make similar plot but for relative change in S (Eq.16) in double logarithmic scale. What do we observe?
4. Set $A_\rho = 1$ to enable the density influencing on the solution. Solve Poisson equation for
 - a) homogeneous boundary conditions using $k_1 = k_2 = k_3 = k_4 = 0$
 - b) inhomogeneous boundary conditions using $k_1 = 1$, $k_2 = -1$, $k_3 = 1$, $k_4 = -1$

In both cases make 2D maps of the potential and error. Now we may easily recognize the regions in which density governs the solution while the BC influences on the potential only a little and vice versa.

5. At home prepare the report, it shall contain short description of the problem and numerical method used in calculations, commented results and conclusions.

3 Example results

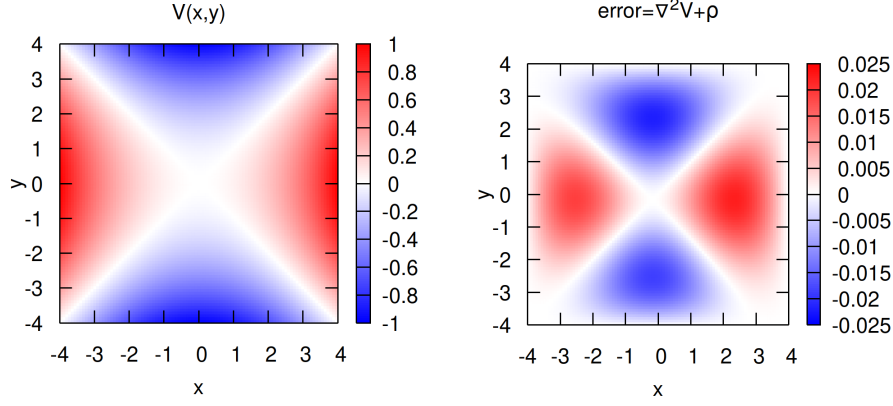


Figure 1: Spatial distribution of the potential (left) and the error (right) for $k_1 = 1$, $k_2 = -1$, $k_3 = 1$, $k_4 = 1$ and $A_\rho = 0$.

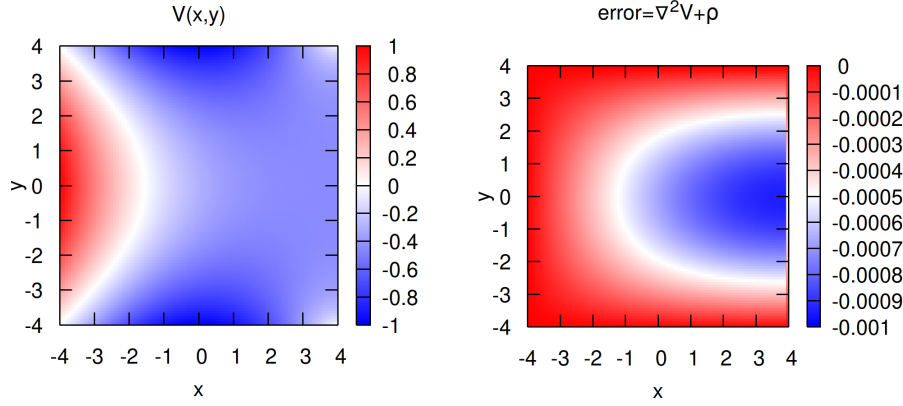


Figure 2: Spatial distribution of the potential (left) and the error (right) for $k_1 = 1$, $k_2 = -1$, $k_3 = 0$, $k_4 = 1$, $A_\rho = 0$ and Neumann BC on the right side $\frac{\partial V}{\partial x} \Big|_{x=x_{max}} = 0$.

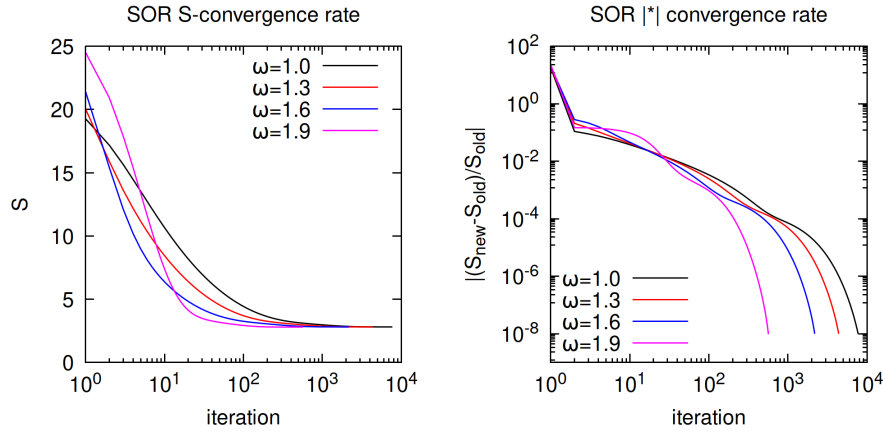


Figure 3: Example convergence of SOR method for a few values of ω parameter: (left) for functional S , and, (right) for the $\delta = |(S_{k+1} - S_k)/S_k|$ in log-log scale.

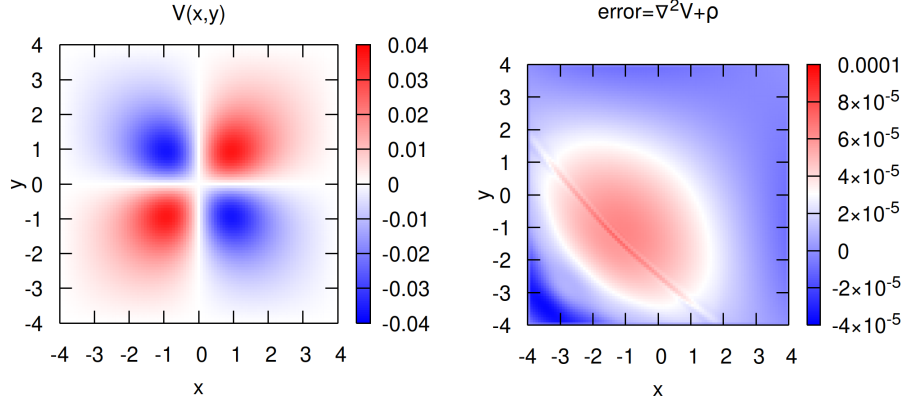


Figure 4: Spatial distribution of the potential (left) and the error (right) for $k_1 = k_2 = k_3 = k_4 = 0$ and $A_\rho = 1$.